Matthew's guidelines for first-time (and later) NSF applicants:

Your **summary page** is the MOST IMPORTANT page of the proposal. All panel reviewers read that page, to decide which proposals they want to review, so that page is your chance to get every panel member excited about your project. Only three panel members will read the rest of the proposal and write reviews.

In general, have a **hypothesis** that you want to prove or disprove. While this is not required, and we do fund exploratory work, having a question you want to answer makes it easier for the panel to figure out what will be learned from your project. It is OKAY if your hypothesis turns out to be wrong (if the answer is obviously "Yes" then there's no reason to do the experiments!), but also write about pitfalls and alternatives if the answer turns out to be "No."

Exploratory work is OK, however, and sometimes necessary. A strategy for exploratory work that can be useful is to start with 1 or 2 or 3 hypothesis-driven aims, and end with a final exploratory aim that can set you up for the grant after this.

It usually helps to have a **VISUAL ABSTRACT** if you can. It does not have to be in the 1-page summary, but such an Abstract really helps the reviewer to quickly understand the gist of your proposal and how the Aims fit together…that's why many top journals request them as well.

Check to MAKE SURE YOUR **AIMS DO NOT DEPEND** ON THE PREVIOUS AIM WORKING! The kiss of death for a proposal is for a reviewer to say "They propose proteomics or binding assays in Aim 1 to find interacting proteins, but what if they don't find any? Then Aim 2 to study those proteins can't be done." This is another reason why an exploratory aim such as a genetic screen or proteomics often is best placed as the last aim.

CDF very much appreciates **computational models**, BUT: Our primary goal is to understand cell biology. If you propose a model for a process, it is much more attractive to reviewers if you plan to test that model experimentally to confirm it. Our reviewers also like it when a proposal follows up experimental work with a computational model. Collaborative efforts between computational and experimental researchers are often very attractive to reviewers.

What are **Broader Impacts**?: I think of these as "Societal Benefits", and they can be broken into 3 components:

A) **Intrinsic applications** of the work you will do. Will this work lead to a better understanding of biology that could be useful to society? Here is where you can mention possible medical implications, or better understandings of cells in general, or advances in understanding crops, or biotechnology advances.

NSF BIO currently emphasizes proposals that address the Bioeconomy, AI, or Quantum effects on biological processes.

B) **Education:** Will you bring trainees (undergrads, grad students, postdocs) into your lab to do research? How will you train them? How will you assess that the training was successful? You will need to provide a Mentoring Plan for grad students and postdocs.

C) **Outreach**: How will you help the general public (or a portion thereof) understand science better? These efforts can involve museum exhibits, making websites, public talks to in public gathering places (e.g. elementary and high schools, 4H clubs, libraries, retirement homes, art galleries, even "Science on Tap" talks in pubs). Grad students and postdocs are often eager to carry out some of these activities, which can be included as part of your mentoring of those trainees.

Reviewers want to see what **extra** you will be doing with your NSF grant that you wouldn't be able to do without these funds. If you just say what you do already, reviewers will comment "They mention the class they teach, but that's what's expected from faculty already." If you already have an NSF or similar grant that is expiring soon, then you can list what you are doing already with that grant but make sure to say that this new grant will allow you to continue that work.

Reviewers will want to see **Assessment** that your activities are effective (just as you assess whether your experiments worked for Intellectual Merit). If a website, how many people visited the site? If an art exhibit, how many people attended? Did you get positive feedback from high school visits? If you invent a game, how many people played it?

You can work with a communications or sociology or educational expert at your institution to help with these, and even ask for some funds in your grant proposal for that person to carry out these assessments.

How long should Broader Impacts be? There is no set length, but you must take them seriously. For proposals that have been funded by CDF, the average length of Broader Impacts is 1½ pages of the 15 pages of the proposal.

NSF does not fund medically motivated research. The questions you are asking must be designed to reveal basic aspects of biology of healthy cells, NOT be designed to treat or understand a disease state. The work can LEAD to medical research (e.g. CDF was the original funder of CRISPR in order to understand how bacteria prevent phage infection), but the goal of THIS proposal must be to clarify how healthy cells work. If you start out your summary telling us how your project is going to cure cancer, we will have to turn down your proposal without reviewing it, and refer you to NIH.

How much money to ask for? There is no set amount to ask for, BUT in recent years, funded proposals asked for **3 or 4 years** of funding at ~$250,000/year, INCLUDING indirect costs. If you ask for $5,000,000, we most likely won't be able to fund your proposal unless it is truly stellar (i.e. your single project is better than 4-6 other strong projects combined!)

Who will review my proposal? Our review panels are not as specialized as those at NIH, and are made up of biologists covering ALL aspects of cell biology. Our panels typically have some people who study archaea and bacteria, some studying invertebrates, at least 3 people who study plants, and a few who study vertebrates. You should try to make sure that all of these can understand your Summary Page! Why is the organism you are using the best choice for studying this topic? If you are using a mammalian cell line, do NOT assume that everyone on the panel will know what that line is; you need to explain why THIS particular cell line is the BEST possible model for answering the question you are asking.

***ad hoc* reviews**: Because the panel may not have 3 members who are expert in your particular field, we may ask outside reviewers who are specialists in your field to review your proposal and send us a written review (much like reviewing a paper). **You can submit names of people who you think are experts in your topic** (in the "Single-Copy Documents" section of your proposal) whose opinion of your proposal you would respect; we encourage these suggestions. You can also submit names of people who you do NOT want to review your proposals (again, in "Single-Copy Documents"; nobody on the panel will see your lists). You will likely receive anywhere from 3-5 reviews for your proposal, 2 or 3 of which were from panelists, the rest as *ad hoc* reviews.

What is the **Panel Summary**? In addition to the panelist reviews, you will receive a "Panel Summary" encapsulating the most important points of discussion that led to the rating of your proposal. NSF does not just average reviewer scores. It is not uncommon to receive an "Excellent" from one panelist who then is swayed by another who found a "fatal flaw" in the proposal that convinced the panel to give your submission a "Low Priority" rating. It is also the case that one panelist makes clear to others that their objections were actually addressed in the proposal and that it is much better than the reviewer scores indicated. Each individual review could have some flaws, but the Panel Summary should list the critical elements that lead to funding, or that must be addressed in a future proposal should it be declined.

Timing of submission: CDF uses a "No-Deadline" model; we set up panels when after receiving at least 30 proposals to review. We generally do not hold panels in the summer (in order to be able to finish fund awards before the end of the federal fiscal year on September 30). Therefore, we often hold panels in early fall, winter, early spring, and late spring, (like many NIH study sections, and for the same reason). Plan on submitting your proposal AT LEAST 6 weeks before those times.

CAREER Award for Assistant Professors:

When to apply: You are eligible to apply for the 5-year CAREER award as long as you are still an Assistant Professor on the CAREER **due date in late July**.

You can only apply for the CAREER award **3 times**. You are strongly encouraged to send a one-page summary of your proposal to a program officer, who can help you determine which program at NSF is most suitable for your ideas. Note that lots of other people are doing the same, AND that the number of program officers was greatly reduced in 2025, so that the earlier you write, the better we can help you.

The goal of the CAREER Award is to help assistant professors develop both as independent researchers AND as educators. You MUST have a strong educational plan (on average, this takes up ~4 pages out of the 15 pages of the proposal). This plan must show what additional educational activities you will undertake in addition to what you are doing now. Instituting a CURE (Course-based Undergraduate Research Experience) class is used in many proposals, but there are many other possibilities. What is important is to show your knowledge of what you will do to train young minds both in the lab, in the classroom, and in society.

Funding levels: Again, about $250,000/year, including indirect costs, is a good goal.

Your chair's letter: Is an important part of the CAREER award. Your chair will of course gush at what a fantastic researcher and wonderful person you are; BUT it really helps if the Chair also describes how they will <u>help you develop</u> as a researcher and educator. They could offer to assign a mentor to help you, or maybe help to fund a piece or equipment if you get the award. Paradoxically, it is sometimes effective if they <u>temporarily REDUCE</u> your teaching load(!) (especially at primarily undergraduate institutions where the teaching load is high) to allow you a bit more time to develop research-based courses. You probably cannot demand your chair do this, but if you have a faculty mentor, that person may be able to make this case to your chair.

Timing of submission: Many (but certainly not all) 1st-year assistant professors just starting after their postdoc do NOT have the educational experience to create a thorough educational plan; after all, many of you are just getting ready to teach your first course! For this reason, I STRONGLY suggest you submit a standard 3-year proposal first, because:

Your CAREER proposal will be competing with more experienced teaching faculty for your educational component.

The Broader Impacts section of a CAREER grant is longer than for a standard grant. Also, as a beginning assistant professor, you may not have as many publications to back up your proposed scientific project, so you will need more room on the proposal to show your preliminary data. This all leaves less room to describe your proposed scientific project; describing a 3-year project is easier than trying to describe a 5-year project.

IF you get funded for the standard project, then….you are funded! And you can still apply for the CAREER award the next year, though the project may need to be different (unless you wait until your first project is almost done before applying for the CAREER).

OR: If you do NOT get funded, then you will get useful experience in writing, get useful feedback to help improve your ideas, and you will still have a full three chances to apply for the CAREER.

Applying for both a CAREER and an NIH proposal? As a beginning investigator, you CAN submit the same research idea to both NSF and another federal agency. HOWEVER, you should STILL tailor your writing to the agency you are writing for. You will still need to write a strong educational section and focus on the basic biology question for the CAREER award; while you can go more deeply into medical aspects for an NIH proposal.